
Problem 1: Fraud Detection System

1 Theoretical Calculation

We use **Bayes' Theorem**

$$P(F|\text{Flag}) = \frac{P(\text{Flag}|F)P(F)}{P(\text{Flag}|F)P(F) + P(\text{Flag}|\neg F)P(\neg F)}$$

Given:

- ($P(F) = 0.02$)
- ($P(\text{Flag}|F) = 0.92$)
- ($P(\text{Flag}|\neg F) = 0.04$)

$$P(F|\text{Flag}) = \frac{0.92 \cdot 0.02}{(0.92 \cdot 0.02) + (0.04 \cdot 0.98)}$$

$$= \frac{0.0184}{0.0184 + 0.0392} = \frac{0.0184}{0.0576} \approx 0.319$$

Final Answer: ~31.9%

2 Why NOT 92%?

Because **92% = $P(\text{Flag} | \text{Fraud})$** , not **$P(\text{Fraud} | \text{Flag})$** .

Key insight:

- Fraud is **rare (2%)**
- Many flagged transactions come from **false positives**

So even a good system produces many wrong alerts.

3 Python Simulation

```
import random

N = 1_000_000

fraud_and_flag = 0
flagged = 0

for _ in range(N):
    is_fraud = random.random() < 0.02

    if is_fraud:
        flagged_txn = random.random() < 0.92
    else:
        flagged_txn = random.random() < 0.04

    if flagged_txn:
        flagged += 1
        if is_fraud:
            fraud_and_flag += 1

print("Estimated P(Fraud | Flagged):", fraud_and_flag / flagged)
```

4 Comparison

- Theoretical: **0.319**
- Simulation: **~0.318–0.321**

Very close → simulation validates theory

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Problem 2: Email Classification

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1 Theoretical Probability

$$P(S|D) = \frac{P(D|S)P(S)}{P(D|S)P(S) + P(D|\neg S)P(\neg S)}$$

Where D = "DISCOUNT"

$$P(S|D) = \frac{0.85 \cdot 0.35}{(0.85 \cdot 0.35) + (0.06 \cdot 0.65)}$$

$$= \frac{0.2975}{0.2975 + 0.039} = \frac{0.2975}{0.3365} \approx 0.884$$

Answer: ~88.4%

2 If spam rate = 10%

$$P(S) = 0.10$$

$$P(S|D) = \frac{0.85 \cdot 0.10}{(0.85 \cdot 0.10) + (0.06 \cdot 0.90)}$$

$$= \frac{0.085}{0.085 + 0.054} = \frac{0.085}{0.139} \approx 0.611$$

New Answer: ~61.1%

Why it changes?

Because of **base rate (prior probability)**:

- When spam is common → DISCOUNT strongly indicates spam
 - When spam is rare → same word becomes less reliable
-

3 Python Simulation

```
import random
```

```
N = 500_000
```

```
spam_and_discount = 0
discount = 0

for _ in range(N):
    is_spam = random.random() < 0.35

    if is_spam:
        has_discount = random.random() < 0.85
    else:
        has_discount = random.random() < 0.06

    if has_discount:
        discount += 1
        if is_spam:
            spam_and_discount += 1

print("Estimated P(Spam | DISCOUNT):", spam_and_discount / discount)
```

4 Role of Base Rate

This is a **base rate fallacy example**:

- Ignoring prior probability leads to wrong conclusions
 - Model performance depends on **data distribution**
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Problem 3: Manufacturing Quality Control

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1 Theoretical Probability

$$P(C|D) = \frac{P(D|C)P(C)}{\sum P(D|Machine)P(Machine)}$$

$$= \frac{0.09 \cdot 0.20}{(0.02 \cdot 0.50) + (0.04 \cdot 0.30) + (0.09 \cdot 0.20)}$$

$$= \frac{0.018}{0.01 + 0.012 + 0.018} = \frac{0.018}{0.04} = 0.45$$

Answer: 45%

2 Which machine causes most defects?

Compute contributions:

- $A \rightarrow 0.50 \times 0.02 = 0.01$
- $B \rightarrow 0.30 \times 0.04 = 0.012$
- $C \rightarrow 0.20 \times 0.09 = 0.018$

Machine C contributes most defects

3 Python Simulation

```
import random
```

```
N = 1_000_000
```

```
defective = 0
```

```
from_C_and_defective = 0
```

```
for _ in range(N):
```

```
    r = random.random()
```

```
    if r < 0.5:
```

```
        machine = 'A'
```

```
        defect = random.random() < 0.02
```

```
    elif r < 0.8:
```

```
        machine = 'B'
```

```
        defect = random.random() < 0.04
```

```
    else:
```

```
        machine = 'C'
```

```
defect = random.random() < 0.09

if defect:
    defective += 1
    if machine == 'C':
        from_C_and_defective += 1

print("Estimated P(C | Defective):", from_C_and_defective / defective)
```

4 Key Insight

Highest defect rate $\neq$ highest defective contribution

Because:

- Machine A produces **more volume**
- Machine C produces **fewer items but worse quality**

Final driver = (**production share $\times$ defect rate**)

Bonus Insight (Convergence)

If you run:

- 10,000 $\rightarrow$ noisy ($\sim \pm 3-5\%$)
- 100,000 $\rightarrow$ closer ($\sim \pm 1\%$)
- 1,000,000 $\rightarrow$ very close ($\sim \pm 0.2\%$)

This happens due to **Law of Large Numbers**
